AI_NAV — Network Route Planning and Flight Control

map Zone1V1 → weighted network → heading-aware A* → route → L1 flight loop

A. Route planning (start and every re-plan)

Mission start
ai_nav_plan(start, end)

Pop-up no-fly zone reported (x, y, r)

blocks remaining route?
- **No** → **keep route**
- **Yes**

1. Resolve start and goal → (x, y, alt)
node name or "x,y"; re-plan: start = aircraft position + course

2. Build the network
every node pair within LINK (or on a corridor):
sample the straight segment every 10 m

inside no-fly + margin or off the sheet?
- **Yes** → **drop link**
- **No**

link weight w = len × corridor mult
$+\sum$(hazard mult − 1) × len in zone
$+ k_{climb}|\Delta\text{alt}|$

3. Heading-aware A* (state = previous node, node)
step cost = $w + k_{turn}|\Delta\psi|$; reject $\Delta\psi > \Delta\psi_{max}$
h = straight distance to goal × min corridor mult

goal reached?
- **No** → **NO ROUTE**
orbit in place, flag set
- **Yes**

4. String-pulling: skip nodes when the line of sight is clear and direct cost ≤ chain cost

5. Publish route[0 . . . n−1] (≤ 32 points)
route_len, route_cost, goal

route[ ]

B. Flight loop (every control cycle)

ai_nav_update(in, dt)
every control cycle

auto_enable?
- **No** → **IDLE, outputs 0**
(manual / RC path)
- **Yes**

state = IDLE?
- **Yes** → **engage**: route[0]
← aircraft position,
leg ← 0, state ← TRACK
- **No**

GPS age update; age > timeout → state ← FAILSAFE

envelope scaling: below $1.15\,V_{min}$ cut the bank limit (to 50 %)
and the pitch-up limit (to 0 at V_{min})

state?

TRACK
L1 carrot on leg $i \to i{+}1$
$\phi = \arctan\frac{2V_g^2 \sin\eta}{L_1 g}$
alt = interpolate waypoint altitudes
turn lead $R_{turn}\tan\frac{\Delta\psi}{2}$: next leg early
last waypoint → LOITER

LOITER
clockwise orbit of the goal
tangent heading + radial correction + centripetal bank feed-forward

FAILSAFE
GPS lost: wings level, hold altitude
(latched until auto is cycled)

$\phi_{cmd} \leftarrow \text{clamp}(\phi_{cmd}, \pm\phi_{max})$

vertical: $\theta_{cmd} = k_{alt}(\text{alt_tgt} - h)$ + bank compensation, limited

throttle: speed PI + climb feed-forward
$V > V_{max} \to 0$; $V < V_{min} \to 1$

inner loops: aileron, elevator = PD on roll, pitch and rates
rudder = $k \sin\phi$ (coordination); flaps = 0

Output → FCS protection layer → servos

return — next control cycle

Legend

- planner step
- flight-loop step
- decision
- data in / out
- hand-over of route[ ]